FINANCIAL STATEMENT ANALYSIS · CASE SET

Context-Based Problems

DuPont Analysis and Altman Z-score

Six short cases with hypothetical data, each built around one learning point

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SOLUTION EDITION · problem, learning objective, formulas, solution, inferences

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Case	Title	Company	Learning point
D1	The Growth That Diluted Returns	Kaveri Appliances Ltd.	Leverage can mask weakening operations: split the change in ROE into its drivers (3-way and 5-way).
D2	Two Roads to the Same ROE	Aurum Jewels Ltd. and Bazaar Fresh Ltd.	Same ROE does not mean same risk: identify the ROE engine and test its sensitivity to a margin squeeze.
D3	The ROE That Nearly Doubled	Sunrise Textiles Ltd.	Headline ROE versus core ROE: strip out one-off gains and their tax effect before judging performance.
Z1	A Safe Borrower Slips	Deccan Auto Components Ltd.	Reading the zones: break the change in the Altman Z-score into its five weighted components.
Z2	When the Share Price Falls Faster than the Business	Marine Chemicals Ltd.	Market-value dependence: how much of Z is the share price, the break-even price, and a book-value cross-check (Z″).
Z3	No Share Price, No Original Z	Gokul Engineering Pvt. Ltd.	Choosing the right model: Z′ for private manufacturers, and why balance-sheet fixes move Z′ less than expected.

How to use this document

- Each case opens with a short context-based problem (about one page), followed by the solution.
- Solutions follow one structure: **learning objective, formulas, step-by-step solution, inferences.**
- All companies, people and data are **hypothetical**. Amounts are in ₹ crore unless stated.

CASE D1 · DUPONT ANALYSIS

The Growth That Diluted Returns

Kaveri Appliances Ltd.

Learning point: Leverage can mask weakening operations: split the change in ROE into its drivers (3-way and 5-way).

Kaveri Appliances, a Pune-based maker of mixers, induction cooktops and water heaters, has just closed its best sales year ever. In FY2026 it opened a second plant at Nashik and added 300 dealers in tier-2 towns. At the results call, the managing director said: *"Profits are up, the business is bigger, and our shareholders are better off."*

Meera Joshi, an equity analyst at Sahyadri Securities, is not convinced. Revenue grew 20%, but profit rose only about 4%. The Nashik plant was funded largely by a new term loan. The board also kept the dividend high, so equity grew by only ₹50 crore. Before she publishes her note, her research head has asked her to explain what really happened to the return earned by shareholders.

The Nashik plant came on stream late in the year, so year-end assets are much higher than the assets that earned most of the year's revenue, and the new loan carries a higher interest rate than the older borrowings. She plans to use the DuPont framework, first in its simple three-way form and then in its fuller five-way form, so that the research head can see the reasons behind the change and not just the headline number.

₹ crore (balances at year-end)	FY2025	FY2026
Revenue	1,200.0	1,440.0
EBIT	168.0	187.2
Interest expense	28.0	41.2
Profit before tax (PBT)	140.0	146.0
Tax expense	35.0	36.5
Profit after tax (PAT)	105.0	109.5
Total assets	800.0	1,080.0
Shareholders' equity	400.0	450.0

Required

1. Compute the **three-way** DuPont ratios (net profit margin, asset turnover, equity multiplier) for both years and verify ROE.
2. Split the net profit margin further and compute the **five-way** ratios: tax burden, interest burden, EBIT margin, asset turnover and equity multiplier.
3. Show how many percentage points each driver added to, or took from, the change in ROE.
4. Evaluate the managing director's claim. Does ROE support it? Is leverage the explanation?
5. In two or three sentences, state what Meera should recommend.

1. Learning objective

Use the DuPont framework to explain *why* ROE changed, separate improvement in operations from improvement through borrowing, and measure each driver's contribution to the change in ROE.

2. Formulas

- **ROE** = PAT ÷ Shareholders' equity
- **Three-way:** ROE = (PAT ÷ Revenue) × (Revenue ÷ Total assets) × (Total assets ÷ Equity) = Net profit margin × Asset turnover × Equity multiplier
- **Five-way:** ROE = (PAT ÷ PBT) × (PBT ÷ EBIT) × (EBIT ÷ Revenue) × (Revenue ÷ Total assets) × (Total assets ÷ Equity) = Tax burden × Interest burden × EBIT margin × Asset turnover × Equity multiplier
- **Net profit margin** = Tax burden × Interest burden × EBIT margin
- **ROA** = PAT ÷ Total assets = Net profit margin × Asset turnover
- **Contribution of a driver (pp)** = ΔROE × ln(driver in FY2026 ÷ driver in FY2025) ÷ ln(ROE in FY2026 ÷ ROE in FY2025). The contributions add up to ΔROE.

3. Solution

Step 1: Three-way decomposition.

Ratio	FY2025	FY2026	Change
Net profit margin	105.0 ÷ 1,200.0 = 8.75%	109.5 ÷ 1,440.0 = 7.60%	−1.15 pp
Asset turnover	1,200.0 ÷ 800.0 = 1.500×	1,440.0 ÷ 1,080.0 = 1.333×	−0.167×
Equity multiplier	800.0 ÷ 400.0 = 2.000×	1,080.0 ÷ 450.0 = 2.400×	+0.400×
ROE (product)	8.75% × 1.500 × 2.000 = **26.25%**	7.60% × 1.333 × 2.400 = **24.33%**	−1.92 pp
Check: PAT ÷ Equity	105.0 ÷ 400.0 = 26.25% ✓	109.5 ÷ 450.0 = 24.33% ✓	

Step 2: Five-way decomposition. The net profit margin is split into tax burden, interest burden and EBIT margin.

Ratio	FY2025	FY2026	Change
Tax burden (PAT ÷ PBT)	105.0 ÷ 140.0 = 0.750	109.5 ÷ 146.0 = 0.750	+0.000
Interest burden (PBT ÷ EBIT)	140.0 ÷ 168.0 = 0.833	146.0 ÷ 187.2 = 0.780	−0.053
EBIT margin (EBIT ÷ Revenue)	168.0 ÷ 1,200.0 = 14.00%	187.2 ÷ 1,440.0 = 13.00%	−1.00 pp
Asset turnover	1.500×	1.333×	−0.167×
Equity multiplier	2.000×	2.400×	+0.400×
ROE (product)	0.750 × 0.833 × 14.00% × 1.500 × 2.000 = **26.25%** ✓	0.750 × 0.780 × 13.00% × 1.333 × 2.400 = **24.33%** ✓	−1.92 pp

Link between the two views: tax burden × interest burden × EBIT margin = 0.750 × 0.780 × 13.00% = 7.60%, which is the net profit margin of Step 1.

Step 3: Contribution of each driver to the change in ROE.

Why logarithms? ROE is a product of ratios, so the changes in the ratios compound and cannot simply be subtracted. A natural logarithm turns a product into a sum: ln(ROE) = ln(margin) + ln(turnover) + ln(multiplier). Each driver's share of the total log change is therefore its share of the change in ROE, and the shares add up exactly.

Formula: Contribution of a driver (pp) = ΔROE × ln(Driver$_2$ ÷ Driver$_1$) ÷ ln(ROE$_2$ ÷ ROE$_1$), with 1 = FY2025 and 2 = FY2026.

3A. Conversion factor (the same for every driver).

Item	Working	Result
ΔROE	24.33% − 26.25%	−1.92 pp
ln(ROE$_2$ ÷ ROE$_1$)	ln(24.33 ÷ 26.25)	−0.0758
Factor = ΔROE ÷ ln(ROE$_2$ ÷ ROE$_1$)	−1.92 ÷ −0.0758	25.28 pp per unit of log

So, for every driver, **Contribution = ln(Driver$_2$ ÷ Driver$_1$) × 25.28**. Full-precision ratios are used; the figures shown are rounded.

3B. Three-way contributions.

Driver	FY2025	FY2026	ln(FY2026 ÷ FY2025)	× factor	Contribution (pp)
Net profit margin	8.75%	7.60%	ln(7.60 ÷ 8.75) = −0.1404	−0.1404 × 25.28	−3.55
Asset turnover	1.500×	1.333×	ln(1.333 ÷ 1.500) = −0.1178	−0.1178 × 25.28	−2.98
Equity multiplier	2.000×	2.400×	ln(2.400 ÷ 2.000) = 0.1823	0.1823 × 25.28	+4.61
Total			**−0.0758**		**−1.92**

3C. Five-way: splitting the margin effect.

Driver	FY2025	FY2026	ln(FY2026 ÷ FY2025)	× factor	Contribution (pp)
Tax burden	0.750	0.750	ln(0.750 ÷ 0.750) = 0.0000	0.0000 × 25.28	+0.00
Interest burden	0.833	0.780	ln(0.780 ÷ 0.833) = −0.0662	−0.0662 × 25.28	−1.67
EBIT margin	14.00%	13.00%	ln(13.00 ÷ 14.00) = −0.0741	−0.0741 × 25.28	−1.87
Margin total			**−0.1404**		**−3.55**

Checks. (1) Because 8.75% = 0.750 × 0.833 × 14.00% (and likewise for FY2026), the logs of the three margin parts add up to the log of the margin: 0.0000 − 0.0662 − 0.0741 = −0.1403, which equals the net profit margin log of −0.1404 in 3B apart from rounding. The five-way margin lines (0.00 − 1.67 − 1.87) therefore add up to −3.55. (2) The three-way contributions add up to ΔROE: −3.55 − 2.98 + 4.61 = −1.92.

How to read it. Margin (−3.55) and asset turnover (−2.98) alone would have cut ROE by 6.53 points. The higher equity multiplier (+4.61) offset most of that, so ROE fell only 1.92 points.

Summary of contributions (pp).

Driver	Three-way (pp)	Five-way (pp)	What happened
Net profit margin	−3.55		Margin shrank from 8.75% to 7.60%
Tax burden		+0.00	Unchanged
Interest burden		−1.67	Interest rose 47% (₹28.0 to ₹41.2 crore)
EBIT margin		−1.87	Costs grew faster than sales
Asset turnover	−2.98	−2.98	Assets +35%, revenue +20%
Equity multiplier	+4.61	+4.61	Assets funded more by borrowing
Total	**−1.92**	**−1.92**	= ROE 26.25% → 24.33%

The three margin lines in the five-way column add up to −3.55, which equals the net profit margin effect in the three-way column (−3.55; difference due to rounding).

Step 4: Test the managing director's claim.

Test	FY2025	FY2026
ROA = PAT ÷ Total assets	105.0 ÷ 800.0 = 13.12%	109.5 ÷ 1,080.0 = 10.14%
ROE if the multiplier stayed at 2.0×	13.13% × 2.0 = 26.25%	10.14% × 2.0 = **20.28%**
Total liabilities (assets − equity)	400.0	630.0
Interest cover (EBIT ÷ interest)	168.0 ÷ 28.0 = 6.0×	187.2 ÷ 41.2 = 4.5×

Step 5: Recommendation. Meera should not endorse the claim. Growth has been bought with lower margins, weaker asset productivity and more debt. In FY2027 she should track whether the Nashik plant lifts asset turnover back towards 1.5× and whether interest cover recovers above 5×.

4. Inferences and answer

- ROE fell from 26.25% to 24.33% *even though* the equity multiplier rose; return on assets fell by nearly 3 pp (13.1% → 10.1%). Leverage masked, rather than explained, the performance.
- The five-way view adds what the three-way view cannot: about half of the margin decline is financing cost (interest burden, −1.67 pp) and half is operating cost (EBIT margin, −1.87 pp).
- Higher ROE from higher leverage is not the same as better performance: shareholders earn less per rupee invested and carry more risk (liabilities up from ₹400 to ₹630 crore).

Answer: ROE: 26.25% → 24.33% (−1.92 pp). Drivers: tax 0.00, interest −1.67, EBIT margin −1.87, asset turnover −2.98, equity multiplier +4.61 (pp).

CASE D2 · DUPONT ANALYSIS

Two Roads to the Same ROE

Aurum Jewels Ltd. and Bazaar Fresh Ltd.

Learning point: Same ROE does not mean same risk: identify the ROE engine and test its sensitivity to a margin squeeze.

Kiran Rao manages a ₹50 crore family-office portfolio and must choose one of two listed consumer companies for a three-year holding. Both reported an ROE of about 18% for FY2026, and a broker note concludes: *"Similar returns, so pick either."* Kiran is not so sure, and has asked an analyst to break each company's return into its parts.

Aurum Jewels sells branded gold and diamond jewellery through 60 large showrooms. Every piece carries a high mark-up, but stock moves slowly and the company is financed mostly by shareholders' money. **Bazaar Fresh** runs 400 neighbourhood supermarkets. Mark-ups are thin, shelves empty within days, and the stores are funded largely by supplier credit and bank loans.

Kiran's economist warns that wage and input-cost inflation could squeeze the operating margin of both businesses by about one percentage point of revenue next year. The two companies pay out a similar share of profit as dividends, so ROE is a fair basis for comparison. The family office's mandate is capital preservation first and growth second, so a company whose earnings are fragile in a bad year would be a poor fit even if it looks equally profitable today. Before committing, Kiran wants to know whether "same ROE" really means "same risk".

FY2026, ₹ crore (balances at year-end)	Aurum Jewels	Bazaar Fresh
Revenue	800.0	4,000.0
EBIT	168.0	100.0
Interest expense	8.0	30.0
Profit before tax (PBT)	160.0	70.0
Tax expense	40.0	17.5
Profit after tax (PAT)	120.0	52.5
Total assets	1,000.0	1,000.0
Shareholders' equity	650.0	285.0

Required

1. Compute the five DuPont ratios for both companies and verify ROE.
2. Classify each company's ROE engine as **margin-led, turnover-led or leverage-led**, giving reasons from the ratios.
3. Assume the EBIT margin falls by one percentage point of revenue, while revenue, interest, the tax rate, assets and equity stay unchanged. Recompute each company's ROE and interest cover (EBIT ÷ interest).
4. Which company would you choose for a risk-averse family office, and why?

1. Learning objective

Show that two companies with the same ROE can have very different risk profiles, by identifying the *engine* of ROE (margin, turnover or leverage) and testing how each engine responds to a margin squeeze.

2. Formulas

- **ROE** = Tax burden × Interest burden × EBIT margin × Asset turnover × Equity multiplier
- **Interest cover** = EBIT ÷ Interest expense
- **Engine test:** the ratio that is most unusual compared with the other company (very high margin → margin-led; very high turnover → turnover-led; very high multiplier → leverage-led).
- **After the squeeze:** EBIT_new = EBIT − 1% × Revenue; PBT_new = EBIT_new − Interest; PAT_new = PBT_new × (1 − tax rate); ROE_new = PAT_new ÷ Equity.

3. Solution

Step 1: Five-way DuPont for both companies (FY2026).

Ratio	Aurum Jewels	Bazaar Fresh
Tax burden	120.0 ÷ 160.0 = 0.750	52.5 ÷ 70.0 = 0.750
Interest burden	160.0 ÷ 168.0 = 0.952	70.0 ÷ 100.0 = 0.700
EBIT margin	168.0 ÷ 800.0 = 21.00%	100.0 ÷ 4,000.0 = 2.50%
Asset turnover	800.0 ÷ 1,000.0 = 0.800×	4,000.0 ÷ 1,000.0 = 4.000×
Equity multiplier	1,000.0 ÷ 650.0 = 1.538×	1,000.0 ÷ 285.0 = 3.509×
ROE (product)	0.750 × 0.952 × 21.00% × 0.800 × 1.538 = **18.46%**	0.750 × 0.700 × 2.50% × 4.000 × 3.509 = **18.42%**
Check: PAT ÷ Equity	120.0 ÷ 650.0 = 18.46% ✓	52.5 ÷ 285.0 = 18.42% ✓

Step 2: Identify the ROE engine.

Company	Standout ratios	Engine
Aurum Jewels	EBIT margin 21.0% (very high); asset turnover 0.80× (low); multiplier 1.54× (low); interest burden 0.952 (almost no interest)	**Margin-led**, little leverage
Bazaar Fresh	EBIT margin 2.5% (very thin); asset turnover 4.00× (very high); multiplier 3.51× (high); interest burden 0.700 (interest takes 30% of EBIT)	**Turnover-led**, helped by high leverage

Step 3: Sensitivity to a one-percentage-point fall in EBIT margin. Revenue, interest, tax rate (25%), assets and equity unchanged.

₹ crore	Aurum (now)	Aurum (squeezed)	Bazaar (now)	Bazaar (squeezed)
EBIT	168.0	160.0 (−8.0)	100.0	60.0 (−40.0)
PBT (EBIT − interest)	160.0	152.0	70.0	30.0
PAT (× 0.75)	120.0	114.0	52.5	22.5
ROE	18.46%	17.54%	18.42%	7.89%
Fall in ROE		−0.92 pp (−5%)		−10.53 pp (−57%)
Interest cover	21.0×	20.0×	3.3×	2.0×

Margin of safety: the EBIT margin at which PBT falls to zero is interest ÷ revenue = 1.0% for Aurum (a fall of 20.0 pp) and 0.75% for Bazaar (a fall of only 1.75 pp).

Step 4: Choice. For a risk-averse family office the choice is **Aurum Jewels**: its ROE rests on a wide margin and equity funding, so it survives a squeeze. Bazaar's identical ROE rests on a very thin margin multiplied by high turnover and leverage, so a small cost shock removes most of its profit.

4. Inferences and answer

- Equal ROE hides unequal risk. Aurum's ROE falls by 5% after the squeeze; Bazaar's falls by 57% and its interest cover drops to 2.0×.
- The DuPont profile, not the ROE number, tells the investor which risks come with the return: operating-cost risk and leverage risk are concentrated in the high-turnover, thin-margin business.

Answer: ROE: Aurum 18.46%, Bazaar 18.42%. After a 1 pp margin squeeze: Aurum 17.54%, Bazaar 7.89%. Choose Aurum.

CASE D3 · DUPONT ANALYSIS

The ROE That Nearly Doubled

Sunrise Textiles Ltd.

Learning point: Headline ROE versus core ROE: strip out one-off gains and their tax effect before judging performance.

Arjun Menon, a fund manager at Tarang Mutual Fund, runs a quarterly screen for companies with improving ROE. This quarter, Sunrise Textiles, a Coimbatore-based maker of cotton yarn and fabric, tops the list: ROE has risen from 12.0% to 20.6%. The press release calls FY2026 "a year of operational excellence".

Arjun reads the annual report and finds three things. First, cotton prices rose sharply and operating margins came under pressure; revenue grew only 3%. Second, the company sold a surplus plot of land for a gain of ₹55 crore, reported as an exceptional item within EBIT, and used the proceeds to repay loans, so total assets and interest both fell. Third, the gain was taxed at 20%, against 30% on regular business profit, so the effective tax rate fell.

Arjun remembers that the same screen once flagged a company whose profit was lifted by a one-off asset sale, and the stock later disappointed. Before adding Sunrise to the fund's buy list, the investment committee wants to know how much of the ROE improvement is real and how much comes from a one-time event.

₹ crore (balances at year-end)	FY2025	FY2026
Revenue	900.0	930.0
Operating profit before exceptional item	81.0	74.4
Exceptional gain on sale of land	0.0	55.0
EBIT (total)	81.0	129.4
Interest expense	21.0	19.4
Profit before tax (PBT)	60.0	110.0
Tax expense (FY2026 includes ₹11.0 on the land gain)	18.0	27.5
Profit after tax (PAT)	42.0	82.5
Total assets	700.0	660.0
Shareholders' equity	350.0	400.0

Required

1. Compute the five-way DuPont ratios for both years using reported figures (EBIT includes the exceptional gain).
2. Identify which ratios drove the rise in ROE and which worked against it.
3. Normalise FY2026: remove the exceptional gain and the tax on it, assume the regular 30% tax rate on the remaining profit, and compute the core EBIT margin, core PAT and core ROE (use closing equity as reported).
4. Compare core ROE with FY2025 ROE and write a three-line note to the committee.

1. Learning objective

Distinguish *headline* ROE from *core* ROE: use the five-way DuPont to see which ratios a one-off gain inflates, then normalise the year to judge underlying performance.

2. Formulas

- **Five-way:** ROE = (PAT ÷ PBT) × (PBT ÷ EBIT) × (EBIT ÷ Revenue) × (Revenue ÷ Total assets) × (Total assets ÷ Equity)
- **Reported EBIT** = Operating profit + Exceptional gain; PBT = EBIT − Interest
- **Core PBT** = Operating profit − Interest; **Core PAT** = Core PBT × (1 − 30%); **Core EBIT margin** = Operating profit ÷ Revenue; **Core ROE** = Core PAT ÷ Equity

3. Solution

Step 1: Reported five-way DuPont.

Ratio	FY2025	FY2026	Change
Tax burden	42.0 ÷ 60.0 = 0.700	82.5 ÷ 110.0 = 0.750	+0.050
Interest burden	60.0 ÷ 81.0 = 0.741	110.0 ÷ 129.4 = 0.850	+0.109
EBIT margin	81.0 ÷ 900.0 = 9.00%	129.4 ÷ 930.0 = 13.91%	+4.91 pp
Asset turnover	900.0 ÷ 700.0 = 1.286×	930.0 ÷ 660.0 = 1.409×	+0.123×
Equity multiplier	700.0 ÷ 350.0 = 2.000×	660.0 ÷ 400.0 = 1.650×	−0.350×
ROE (product)	0.700 × 0.741 × 9.00% × 1.286 × 2.000 = **12.00%**	0.750 × 0.850 × 13.91% × 1.409 × 1.650 = **20.63%**	+8.63 pp
Check: PAT ÷ Equity	42.0 ÷ 350.0 = 12.00% ✓	82.5 ÷ 400.0 = 20.63% ✓	

Step 2: Which ratios drove the rise (contribution to ΔROE = +8.63 pp).

Formula: Contribution of a driver (pp) = ΔROE × ln(Driver$_2$ ÷ Driver$_1$) ÷ ln(ROE$_2$ ÷ ROE$_1$), with 1 = FY2025 and 2 = FY2026. The logarithm turns the product of the five ratios into a sum, so the five contributions add up exactly to ΔROE.

2A. Conversion factor (the same for every driver).

Item	Working	Result
ΔROE	20.63% − 12.00%	+8.63 pp
ln(ROE$_2$ ÷ ROE$_1$)	ln(20.63 ÷ 12.00)	0.5416
Factor = ΔROE ÷ ln(ROE$_2$ ÷ ROE$_1$)	+8.63 ÷ 0.5416	15.93 pp per unit of log

So **Contribution = ln(Driver$_2$ ÷ Driver$_1$) × 15.93**.

2B. Contribution of each driver.

Driver	FY2025	FY2026	ln(FY2026 ÷ FY2025)	× factor	Contribution (pp)
Tax burden	0.700	0.750	ln(0.750 ÷ 0.700) = 0.0690	0.0690 × 15.93	+1.10
Interest burden	0.741	0.850	ln(0.850 ÷ 0.741) = 0.1377	0.1377 × 15.93	+2.19
EBIT margin	9.00%	13.91%	ln(13.91 ÷ 9.00) = 0.4357	0.4357 × 15.93	+6.94
Asset turnover	1.286×	1.409×	ln(1.409 ÷ 1.286) = 0.0916	0.0916 × 15.93	+1.46
Equity multiplier	2.000×	1.650×	ln(1.650 ÷ 2.000) = −0.1924	−0.1924 × 15.93	−3.06
Total			**0.5416**		**+8.63**

Check. 1.10 + 2.19 + 6.94 + 1.46 − 3.06 = +8.63 = 20.63% − 12.00%. The reported ROE ratios multiply back to ROE, so the logs add back to ln(ROE$_2$ ÷ ROE$_1$) = 0.5416 (0.0690 + 0.1377 + 0.4357 + 0.0916 − 0.1924).

Summary and comments.

Driver	Contribution (pp)	Comment
Tax burden	+1.10	Land gain taxed at 20%, not 30%
Interest burden	+2.19	Gain adds to EBIT; loans repaid, interest falls
EBIT margin	+6.94	**Largest:** the ₹55 crore gain lifts EBIT from 8.0% to 13.9% of revenue
Asset turnover	+1.46	Revenue +3% on 6% lower assets (land sold)
Equity multiplier	−3.06	Lower leverage works against ROE (loans repaid)
Total	**+8.63**	= 12.00% → 20.63%

Step 3: Normalise FY2026 (remove the ₹55 crore gain and the ₹11.0 crore tax on it).

₹ crore	FY2026 reported	Adjustment	FY2026 core
Operating profit (core EBIT)	74.4 + 55.0 = 129.4	− 55.0	**74.4**
Interest	19.4	–	19.4
PBT	110.0	− 55.0	**55.0**
Tax	27.5	16.5 at 30% (− 11.0)	**16.5**
PAT	82.5	− 44.0	**38.5**
EBIT margin	13.91%		74.4 ÷ 930.0 = **8.00%**
ROE on closing equity ₹400.0	20.63%		38.5 ÷ 400.0 = **9.63%**
ROE if the retained after-tax gain is removed from equity (400.0 − 44.0 = 356.0)			38.5 ÷ 356.0 = **10.81%**

Step 4: Note to the committee. (1) Reported ROE rose from 12.0% to 20.6% only because of a one-time ₹55 crore land gain. (2) Core ROE fell from 12.0% to 9.6% (10.8% at best) as cotton costs cut the core margin from 9.0% to 8.0%. (3) Do not add the stock on the screen; revisit when core margins recover.

4. Inferences and answer

- The headline ROE nearly doubled; the underlying ROE fell by 2.4 to 1.2 pp. A single exceptional item moved EBIT margin, interest burden and tax burden at once.
- Always compute ROE from *recurring* profit before ranking companies. Screens based on reported ROE reward one-off gains.

Answer: Reported ROE 12.00% → 20.63%. Core FY2026: EBIT margin 8.00%, PAT ₹38.5 crore, ROE 9.63% (10.81% on reduced equity).

CASE Z1 · ALTMAN Z-SCORE

A Safe Borrower Slips

Deccan Auto Components Ltd.

Learning point: Reading the zones: break the change in the Altman Z-score into its five weighted components.

Anita Desai is a credit analyst at Mahanadi Bank. Every year she reviews the ₹150 crore working-capital limit extended to Deccan Auto Components, a supplier of brake and clutch parts to passenger-car makers. FY2026 was difficult: two large customers cut production schedules in the second half, sales fell 4%, finished-goods stock piled up, and the company drew more short-term loans to fund the inventory. The share price also fell as investors marked down the auto-ancillary sector. The promoters have asked for the limit to be renewed unchanged, and Anita knows that auto-component makers are capital-intensive and cyclical, and that she must justify in writing any change in the bank's exposure.

The bank's policy is to compute the Altman Z-score at every review: **Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5**, where X1 = working capital ÷ total assets, X2 = retained earnings ÷ total assets, X3 = EBIT ÷ total assets, X4 = market value of equity ÷ total liabilities and X5 = sales ÷ total assets. A score above 2.99 is *safe*, 1.81 to 2.99 *grey* and below 1.81 *distress*. A move from one zone to another is referred to the credit committee, which will want to know not just the score but the reasons for any change. Last year the company scored comfortably in the safe zone.

₹ crore (year-end)	FY2025	FY2026
Current assets	520.0	560.0
Current liabilities	300.0	430.0
Total assets	1,000.0	1,150.0
Retained earnings	380.0	395.0
EBIT	150.0	92.0
Market capitalisation	1,100.0	640.0
Total liabilities	520.0	700.0
Sales	1,300.0	1,250.0

Required

1. Compute X1 to X5 and the Z-score for FY2025 and FY2026, and state the zone for each year.
2. Compute the weighted contribution of each term (1.2X1, 1.4X2, 3.3X3, 0.6X4, 1.0X5) and identify which term explains most of the change in Z.
3. Holding everything else constant, by how much would EBIT have to rise to bring FY2026 Z back to 2.99? Is that realistic?
4. State Anita's recommendation to the credit committee in three lines.

1. Learning objective

Interpret the Altman zones and identify *which component* of the score explains a change in Z, so that a credit committee is told the reasons, not just the number.

2. Formulas

- **Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5**
- X1 = (Current assets − Current liabilities) ÷ Total assets; X2 = Retained earnings ÷ Total assets; X3 = EBIT ÷ Total assets
- X4 = Market capitalisation ÷ Total liabilities; X5 = Sales ÷ Total assets
- **Zones:** $Z > 2.99$ safe; $1.81 \le Z \le 2.99$ grey; $Z < 1.81$ distress
- **Required EBIT rise to reach 2.99:** $\Delta X3 = (2.99 - Z) \div 3.3$; $\Delta EBIT = \Delta X3 \times$ Total assets

3. Solution

Step 1: The five ratios.

Ratio	FY2025	FY2026
X1 = (CA − CL) ÷ TA	(520.0 − 300.0) ÷ 1,000.0 = 0.2200	(560.0 − 430.0) ÷ 1,150.0 = 0.1130
X2 = RE ÷ TA	380.0 ÷ 1,000.0 = 0.3800	395.0 ÷ 1,150.0 = 0.3435
X3 = EBIT ÷ TA	150.0 ÷ 1,000.0 = 0.1500	92.0 ÷ 1,150.0 = 0.0800
X4 = Mkt cap ÷ TL	1,100.0 ÷ 520.0 = 2.1154	640.0 ÷ 700.0 = 0.9143
X5 = Sales ÷ TA	1,300.0 ÷ 1,000.0 = 1.3000	1,250.0 ÷ 1,150.0 = 1.0870

Step 2: Weighted terms, Z-score and zone; contribution of each term to the change.

Formula: Share of fall = Change in the term ÷ Total change in Z, where Change in the term = FY2026 term − FY2025 term, and Total change in Z = the sum of the five changes. The shares add up to 100%.

Term	FY2025	FY2026	Change	Share of fall
1.2·X1	0.2640	0.1357	−0.1283	9.55%
1.4·X2	0.5320	0.4809	−0.0511	3.80%
3.3·X3	0.4950	0.2640	−0.2310	17.19%
0.6·X4	1.2692	0.5486	−0.7207	53.61%
1.0·X5	1.3000	1.0870	−0.2130	15.85%
Z-score	**3.86**	**2.52**	**−1.34**	100.00%
Zone	**Safe** (> 2.99)	**Grey** (1.81 to 2.99)	Zone change	

How the shares are calculated. Total change in Z = 2.5160 − 3.8602 = −1.3442 (shown as −1.34 in the table). Each term's share is its own change divided by this total.

Term	Working	Change ÷ Total	Share of fall
1.2·X1	0.1357 − 0.2640 = −0.1283	−0.1283 ÷ −1.3442	9.55%
1.4·X2	0.4809 − 0.5320 = −0.0511	−0.0511 ÷ −1.3442	3.80%
3.3·X3	0.2640 − 0.4950 = −0.2310	−0.2310 ÷ −1.3442	17.19%
0.6·X4	0.5486 − 1.2692 = −0.7207	−0.7207 ÷ −1.3442	53.61%
1.0·X5	1.0870 − 1.3000 = −0.2130	−0.2130 ÷ −1.3442	15.85%
Z-score	**2.5160 − 3.8602 (sum of the five changes)**	**−1.3442 ÷ −1.3442**	**100.00%**

The market-value term (0.6X4) explains 54% of the fall; EBIT-to-assets (3.3X3) and sales-to-assets (1.0X5) explain 17% and 16%.

Step 3: How much EBIT is needed to restore Z to 2.99?

Item	Working
Gap to safe zone	2.99 − 2.516 = 0.474
Required rise in X3	0.474 ÷ 3.3 = 0.1436
Required rise in EBIT	0.1436 × 1,150.0 = **₹165.2 crore**
Required EBIT	92.0 + 165.2 = ₹257.2 crore, i.e. 2.8× the FY2026 EBIT and 1.7× the FY2025 EBIT

Step 4: Recommendation (three lines). (1) The borrower has moved from the safe zone to the grey zone: Z fell from 3.86 to 2.52, mainly because market value fell, and also because earning power and asset turnover weakened. (2) EBIT alone cannot restore the safe zone, so renewal should rest on a recovery plan rather than on the score. (3) Renew the limit with enhanced monitoring: quarterly Z-score, inventory and receivable reviews, and a drawing-power cap linked to stock.

4. Inferences and answer

- Z fell by 1.34, and more than half of the fall is the share price (X4). A grey-zone score driven by market value is a warning of lower confidence, not proof of insolvency; the operating weakening (X3, X5) is the part the bank can act on.
- Reading Z by components tells the committee *why*: weaker earning power (X3), slower asset turnover (X5), and a thinner working-capital cushion (X1) alongside the falling market value.

Answer: Z: 3.86 (Safe) → 2.52 (Grey). 0.6X4 explains 54% of the fall. EBIT would need to rise by about ₹165 crore.

CASE Z2 · ALTMAN Z-SCORE

When the Share Price Falls Faster than the Business

Marine Chemicals Ltd.

Learning point: Market-value dependence: how much of Z is the share price, the break-even price, and a book-value cross-check (Z").

Marine Chemicals, a listed maker of industrial solvents, is a long-standing borrower of Coastal Finance Bank. Three weeks ago a gas leak at its main plant caused a shutdown, a regulatory notice and a social-media storm. The share price fell from ₹90 to ₹38 in five trading days. The latest audited balance sheet, which pre-dates the incident, shows no deterioration, and management insists that "the business fundamentals are unchanged". The bank has ₹120 crore outstanding to the company, secured on plant and inventory. A wrongly triggered watch-list entry could damage a fifteen-year relationship; a missed warning could prove costly.

Rahul Iyer, the bank's risk manager, must tell the risk committee tomorrow whether to move the borrower to the watch list. The early-warning model uses the Altman Z-score (**Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5**, with X4 = market value of equity ÷ total liabilities; safe above 2.99, distress below 1.81). Rahul wonders how much of any change in Z comes simply from the price and not from the business. He also has a book-value variant that ignores the share price: **Z″ = 6.56X1 + 3.26X2 + 6.72X3 + 1.05X4**, with X4 = book equity ÷ total liabilities (safe above 2.60, distress below 1.10).

₹ crore unless stated	Latest balance sheet
Current assets	640.0
Current liabilities	410.0
Total assets	1,600.0
Retained earnings	520.0
EBIT	190.0
Total liabilities	900.0
Sales	1,500.0
Shares outstanding (crore)	20.0
Share price before / after the incident (₹)	90 / 38

Required

1. Compute the Z-score at ₹90 and at ₹38, and state the zone in each case.
2. Find the lowest share price at which the company still scores *safe*. Can the company reach the distress zone at any share price?
3. Compute Z″ from book values and compare it with the original Z.
4. Advise the committee: should the borrower go on the watch list, and how should the bank use the two scores together?

1. Learning objective

Measure how much of the Altman Z-score is the *share price*, find the break-even price for the safe zone, and use a book-value variant (Z″) as a cross-check before acting on a market-driven signal.

2. Formulas

- **Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5**, X4 = Market capitalisation ÷ Total liabilities (market capitalisation = price × shares)
- **Break-even for 2.99:** 0.6 × X4 = 2.99 − (1.2X1 + 1.4X2 + 3.3X3 + 1.0X5); Market cap = X4 × Total liabilities; Price = Market cap ÷ Shares
- **Z″ = 6.56X1 + 3.26X2 + 6.72X3 + 1.05X4**, X4 = Book equity ÷ Total liabilities; safe > 2.60, distress < 1.10

3. Solution

Step 1: Z at ₹90 and at ₹38. X1, X2, X3 and X5 do not depend on the price; only X4 changes.

Term	Working	At ₹90	At ₹38
X1 = (640 − 410) ÷ 1,600		0.1437	0.1437
X2 = 520 ÷ 1,600		0.3250	0.3250
X3 = 190 ÷ 1,600		0.1188	0.1188
Market capitalisation	90 × 20 = 1,800; 38 × 20 = 760	1,800.0	760.0
X4 = Mkt cap ÷ 900	1,800 ÷ 900; 760 ÷ 900	2.0000	0.8444
X5 = 1,500 ÷ 1,600		0.9375	0.9375
1.2X1 + 1.4X2 + 3.3X3 + 1.0X5	price-independent part	1.9569	1.9569
0.6X4		1.2000	0.5067
Z-score		**3.16**	**2.46**
Zone		**Safe**	**Grey**

A 58% fall in price (₹90 → ₹38) moves Z by 0.69 and the zone from safe to grey, with no change in the accounts.

Step 2: Break-even price for the safe zone; can distress be reached?

Item	Working
Price-independent part of Z	1.9569
0.6X4 needed for Z = 2.99	2.99 − 1.9569 = 1.0331
X4 needed	1.0331 ÷ 0.6 = 1.7219
Market capitalisation needed	1.7219 × 900 = ₹1,549.7 crore
Lowest "safe" share price	1,549.7 ÷ 20 = **₹77.48** (a fall of only 13.9% from ₹90)
Lowest possible Z	At market cap = 0, Z = 1.9569, still above 1.81: **distress is unreachable on price alone**

Step 3: Book-value cross-check (Z″). Book equity = 1,600 − 900 = 700, so X4 = 700 ÷ 900 = 0.7778.

Term	Working	Value
6.56X1	6.56 × 0.14375	0.9430
3.26X2	3.26 × 0.3250	1.0595
6.72X3	6.72 × 0.11875	0.7980
1.05X4	1.05 × 0.7778	0.8167
Z″		**3.62** (Safe, above 2.60)

Step 4: Advice to the committee. Put the borrower on the watch list for *monitoring*: the market signals higher risk and the shutdown costs are not yet in the accounts. Do not downgrade the credit yet: the fall in Z is entirely price-driven, Z″ is safe at 3.62, and distress is out of reach on price alone. Review when post-incident results are published; use Z as the early warning and Z″ as the check on fundamentals.

4. Inferences and answer

- About 0.70 of Z (3.16 to 2.46) is the share price alone: the score is fast but noisy as an early-warning tool.
- A market-based and a book-based score answer different questions: use the first to flag, the second to verify.

Answer: Z: 3.16 (Safe) at ₹90; 2.46 (Grey) at ₹38. Safe-zone break-even price ≈ ₹77.48. Z cannot fall below 1.96. Z″ = 3.62 (Safe).

CASE Z3 · ALTMAN Z-SCORE

No Share Price, No Original Z

Gokul Engineering Pvt. Ltd.

Learning point: Choosing the right model: Z' for private manufacturers, and why balance-sheet fixes move Z' less than expected.

Gokul Engineering, a family-owned maker of precision castings, has applied to Western Cooperative Bank for a ₹100 crore term loan to modernise its foundry. The promoters hold all the shares and have never raised outside equity, and the largest customer accounts for 30% of sales. The company is unlisted, so its shares have no market price. Priya Nair, the bank's credit officer, therefore applies the private-company version of the Altman model prescribed in the bank's credit manual: **$Z' = 0.717X1 + 0.847X2 + 3.107X3 + 0.420X4 + 0.998X5$**, where X4 = book equity ÷ total liabilities and the other ratios are defined as in the original model. Zones: above 2.90 safe; 1.23 to 2.90 grey; below 1.23 distress.

The first score disappoints. The promoters, keen to win the loan, put forward three proposals: **(A)** convert ₹120 crore of existing long-term loans from the family into equity; **(B)** sell idle land and machinery with a book value of ₹80 crore and use the proceeds to repay term loans (these assets earn no profit); **(C)** cut costs to add ₹20 crore to EBIT, with no change in the balance sheet. The bank's policy requires a safe-zone score for unsecured lending, while a grey-zone score calls for extra security and covenants. Priya must advise which proposals the bank should insist on.

₹ crore (year-end)	Gokul Engineering
Current assets	300.0
Current liabilities	260.0
Total assets	800.0
Retained earnings	60.0
EBIT	70.0
Total liabilities	560.0
Book equity (total assets − total liabilities)	240.0
Sales	960.0

Required

1. Explain in two lines why the original Z cannot be used here. Compute Z′ and state the zone.
2. Recompute Z′ under proposals A, B and C separately. Which gives the largest improvement?
3. Compute Z′ if all three are implemented together. Does the company reach the safe zone?
4. What should Priya conclude about the loan application?

1. Learning objective

Choose the right Altman model for the borrower (listed or unlisted; manufacturer or not), and learn which balance-sheet and operating changes move the score the most.

2. Formulas

- **Z′ = 0.717X1 + 0.847X2 + 3.107X3 + 0.420X4 + 0.998X5** (private manufacturing firms)
- X1 = (CA − CL) ÷ TA; X2 = RE ÷ TA; X3 = EBIT ÷ TA; **X4 = Book equity ÷ Total liabilities**; X5 = Sales ÷ TA
- **Zones:** Z′ > 2.90 safe; 1.23 ≤ Z′ ≤ 2.90 grey; Z′ < 1.23 distress

3. Solution

Step 1: Model choice and the base score. The original Z needs the market value of equity for X4. Gokul is unlisted, so no market price exists; the private-company version Z′ replaces it with book equity ÷ total liabilities.

Term	Working	Ratio	Weighted
X1	(300.0 − 260.0) ÷ 800.0	0.0500	0.717 × = 0.0358
X2	60.0 ÷ 800.0	0.0750	0.847 × = 0.0635
X3	70.0 ÷ 800.0	0.0875	3.107 × = 0.2719
X4	240.0 ÷ 560.0	0.4286	0.420 × = 0.1800
X5	960.0 ÷ 800.0	1.2000	0.998 × = 1.1976
Z′			**1.75** (Grey: 1.23 to 2.90)

Step 2: Each proposal separately.

Proposal	Change in inputs	X1 | X2 | X3 | X4 | X5	Z′	Gain
Base	–	0.050 | 0.075 | 0.087 | 0.429 | 1.200	1.75	–
A: loans → equity	TL 560 → 440; Equity 240 → 360	0.050 | 0.075 | 0.087 | 0.818 | 1.200	1.91	+0.16
B: sell idle assets, repay loans	TA 800 → 720; TL 560 → 480	0.056 | 0.083 | 0.097 | 0.500 | 1.333	1.95	+0.20
C: EBIT + ₹20 crore	EBIT 70 → 90	0.050 | 0.075 | 0.113 | 0.429 | 1.200	1.83	+0.08

Where the gain comes from: in A, only X4 changes (0.429 → 0.818), worth 0.420 × 0.390 = +0.16. In B, most of the gain (+0.13 of +0.20) is higher asset turnover (X5) because idle assets leave the denominator, and the rest is X3, X4, X1 and X2. In C, X3 rises by 0.025, worth 3.107 × 0.025 = +0.08.

Step 3: All three together (TA 720; TL 320 = 480 − 120; equity 400; EBIT 90).

Item	Value
X1 | X2 | X3 | X4 | X5	0.056 | 0.083 | 0.125 | 1.250 | 1.333
Z′	**2.35** (Grey; still below 2.90 by 0.55)

Step 4: Conclusion for the loan. Gokul is in the grey zone today and stays there even after all three measures. B delivers the largest single improvement, but the bigger lesson is that balance-sheet engineering alone cannot move the score into the safe zone. Priya should recommend approval only with security, covenants and staged disbursement linked to operating improvement (EBIT, asset turnover), and ask for the debt-to-equity conversion as a condition.

4. Inferences and answer

- Z′ gives X4 a low weight (0.42), so recapitalising barely moves it (+0.16). Earning power (X3, weight

3.1) and asset productivity (X5) matter more, which means operating improvement beats financial restructuring.

- Using the original Z for an unlisted firm would be impossible (no market price) and using a listed-firm proxy would be misleading. Choosing the right model variant comes first.

Answer: $Z' = 1.75$ (Grey). A: 1.91, B: 1.95, C: 1.83, A + B + C: 2.35 (Grey). B gives the largest improvement; none reaches 2.90.